

LAPORAN PRAKTIKUM 10



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Mata kuliah : Pemrograman Web

Tujuan Praktikum

Setelah mengikuti praktikum ini mahasiswa diharapkan dapat:

1. Memahami konsep dasar JavaScript sebagai bahasa pemrograman web.
2. Memahami fitur Modern JavaScript (ES6+) dan perannya sebagai fondasi React.
3. Mampu menggunakan sintaks ES6+ secara tepat dan konsisten.
4. Memahami konsep state, immutability, dan functional programming.
5. Mampu membangun aplikasi sederhana berbasis state menggunakan Vanilla JavaScript.

Untuk melaksanakan praktikum ini, diperlukan:

Untuk menjalankan modul ini, diperlukan komputer atau laptop dengan sistem operasi Windows, Linux, atau macOS, minimal prosesor 1 GHz, RAM 2GB (disarankan 4GB atau lebih), serta browser modern seperti Google Chrome, Mozilla Firefox, atau Microsoft Edge. Perangkat lunak yang digunakan meliputi:

- Code editor (Visual Studio Code, Sublime Text, atau sejenisnya)
- Web browser

Bahan yang dibutuhkan:

- File HTML dasar
- File JavaScript (.js)

Langkah-Langkah Praktikum

Index.html

```
<!doctype html>
<html lang="id">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
    <title>Vanilla JS Web Store</title>
```

```

    <meta name="description" content="Toko elektronik
modern berbasis Vanilla JavaScript ES6+" />
    <link rel="preconnect"
href="https://fonts.googleapis.com" />
    <link rel="preconnect"
href="https://fonts.gstatic.com" crossorigin />
    <link
        href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&family=Merriweather:ital,wght@0,400;0,700;1,400&display=swap"
        rel="stylesheet"
    />
    <link rel="stylesheet" href="style.css" />
    <script src="https://code.jquery.com/jquery-
3.7.1.min.js"></script>
</head>

<body>
    <div id="app">
        <div class="container">

            <div class="listing-header">
                <nav class="breadcrumb" aria-
label="Breadcrumb">
                    <a href="#">
                        <svg xmlns="http://www.w3.org/2000/svg"
fill="currentColor" viewBox="0 0 20 20">
                            <path d="m19.707 9.293-2-2-7-7a1 1 0 0
0-1.414 0l-7 7-2 2a1 1 0 0 0 1.414 1.414L2 10.414V18a2
2 0 0 0 2 2h3a1 1 0 0 0 1-1v-4a1 1 0 0 1 1-1h2a1 1 0 0
1 1 1v4a1 1 0 0 0 1 1h3a2 2 0 0 0 2-2v-7.586l.293.293a1
1 0 0 0 1.414-1.414Z" />
                        </svg>
                        Home
                    </a>
                    <svg style="color: #9CA3AF;"
xmlns="http://www.w3.org/2000/svg" fill="none"
viewBox="0 0 24 24">

```

```

        <path stroke="currentColor" stroke-
linecap="round" stroke-linejoin="round" stroke-
width="2" d="m9 5 7 7-7 7" />
    </svg>
    <a href="#">Products</a>
    <svg style="color: #9CA3AF;"
xmlns="http://www.w3.org/2000/svg" fill="none"
viewBox="0 0 24 24">
        <path stroke="currentColor" stroke-
linecap="round" stroke-linejoin="round" stroke-
width="2" d="m9 5 7 7-7 7" />
    </svg>
    <span style="color: var(--text-
muted);">Electronics</span>
</nav>

<div class="toolbar">
    <h2 class="page-title">Electronics</h2>
    <div class="toolbar-actions">
        <button class="btn-secondary" id="btn-
filter">
            <svg xmlns="http://www.w3.org/2000/svg"
fill="none" viewBox="0 0 24 24">
                <path stroke="currentColor" stroke-
linecap="round" stroke-width="2" d="M18.796 4H5.204a1 1
0 0 0-.753 1.659l5.302 6.058a1 1 0 0 1
.247.659v4.874a.5.5 0 0 0 .24l3 2.25a.5.5 0 0 0 .8-
.4v-7.124a1 1 0 0 1 .247-.659l5.302-6.059c.566-
.646.106-1.658-.753-1.658Z" />
            </svg>
            Filters
        </button>
        <button class="btn-secondary" id="btn-
sort">
            <svg xmlns="http://www.w3.org/2000/svg"
fill="none" viewBox="0 0 24 24">
                <path stroke="currentColor" stroke-
linecap="round" stroke-linejoin="round" stroke-

```

```

width="2" d="M7 4v16M7 4l3 3M7 4 4 7m9-3h6l-6 6h6m-6.5
10 3.5-7 3.5 7M14 18h4" />
    </svg>
    Sort
  </button>
  <!-- Cart toggle button -->
  <button class="btn-secondary" id="btn-
cart-toggle">
    <svg xmlns="http://www.w3.org/2000/svg"
fill="none" viewBox="0 0 24 24">
      <path stroke="currentColor" stroke-
linecap="round" stroke-linejoin="round" stroke-
width="2" d="M4 4h1.5L8 16m0 0h8m-8 0a2 2 0 1 0 0 4 2 2
0 0 0 0-4Zm8 0a2 2 0 1 0 0 4 2 2 0 0 0 0-4Zm.75-
3H7.5M11 7H6.312M17 4v6m-3-3h6" />
    </svg>
    Cart (<span id="cart-count">0</span>)
  </button>
</div>
</div>
</div>
</div>

  <div id="product-list" class="product-
grid"></div>

  <div id="cart" class="cart-section"></div>

</div>
</div>

<!-- Cart Sidebar -->
<div id="cart-sidebar">
  <h2>Your Cart 🛒</h2>
  <div class="cart-list" id="sidebar-cart-
list"></div>
  <div class="cart-footer-summary" id="sidebar-
cart-footer"></div>
</div>

```

```
<!-- Overlay -->
<div id="overlay" onclick="closeSidebar()"></div>

<script type="module" src="app.js"></script>
</body>
</html>
```

App.js

```
import { fetchProducts } from
"./src/api/productApi.js";
import { getState, setState } from
"./src/core/Store.js";
import { ProductList } from
"./src/components/ProductList.js";
import { Cart } from "./src/components/Cart.js";
import { Newsletter } from
"./src/components/Newsletter.js";

function addToCart(product) {
  const state = getState();

  const exists = state.cart.find((item) => item.id ===
product.id);
  if (exists) {
    showToast(`${product.title}` sudah ada di
keranjang!`);
    return;
  }
}
```

```
    setState({
      cart: [...state.cart, product],
    });

    updateCartCount();
    render();
    showToast(`✅ "${product.title}" ditambahkan ke
keranjang!`);
  }

// Fungsi: Checkout

function checkout() {
  setState({ cart: [] });
  updateCartCount();
  closeSidebar();
  render();
  showToast("👋 Terima kasih atas pembelian Anda!");
}

// Fungsi: Render ulang semua komponen

function render() {
  const state = getState();
  ProductList(state.products, addToCart);
  Cart(state.cart, checkout);
}

// Fungsi: Update badge jumlah item di cart

function updateCartCount() {
  const state = getState();
  $("#cart-count").text(state.cart.length);
}
```

```
// Fungsi: Tampilkan sidebar cart

function openSidebar() {
  $("#cart-sidebar").addClass("open");
  $("#overlay").addClass("show");
}

// Fungsi global agar bisa dipanggil dari HTML onclick
window.closeSidebar = function () {
  $("#cart-sidebar").removeClass("open");
  $("#overlay").removeClass("show");
};

// Fungsi: Toast notification (jQuery)

function showToast(message) {
  // Buat atau ambil elemen toast
  let $toast = $(".toast");
  if (!$toast.length) {
    $toast = $('<div class="toast"></div>');
    $("body").append($toast);
  }

  $toast.text(message).addClass("show");

  setTimeout(() => {
    $toast.removeClass("show");
  }, 2500);
}

// Fungsi: Inisialisasi aplikasi

async function init() {
  try {
    // Tampilkan skeleton loading
    $("#product-list").html(
      Array(6)
```



```

        .fill(
            `

style.css


```

```

/*
=====
=====
style.css - Vanilla JS Web Store
Modul 10: Modern JavaScript ES6+
=====
===== */

:root {
  --primary: #0D9488;
  --primary-hover: #0F766E;
  --primary-light: #CCFBF1;
  --bg-color: #F9FAFB;
  --card-bg: #FFFFFF;
  --text-main: #111827;
  --text-muted: #6B7280;
  --border-color: #E5E7EB;
  --rating-star: #FACC15;
  --font-serif: 'Merriweather', serif;
  --font-sans: 'Inter', sans-serif;
  --radius: 8px;
}

/* Reset & Base */
* { box-sizing: border-box; margin: 0; padding: 0; }
body {
  font-family: var(--font-sans);
  background: var(--bg-color);
  color: var(--text-main);
  line-height: 1.5;
  -webkit-font-smoothing: antialiased;
}

/* Layout */
.container { max-width: 1280px; margin: 0 auto;
padding: 32px 16px; }
.flex-row, .breadcrumb, .breadcrumb a, .toolbar,
.toolbar-actions,

```

```
.feature, .rating-row, .card-footer { display: flex;
align-items: center; }

/* Buttons */
.btn-primary, .btn-secondary {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 8px;
  border-radius: var(--radius);
  font-size: 0.875rem;
  font-weight: 500;
  cursor: pointer;
  transition: 0.2s;
}
svg { width: 16px; height: 16px; }

/* Header */
.listing-header { margin-bottom: 32px; }
.page-title { font-family: var(--font-serif); font-size: 1.875rem; font-weight: 700; }
.toolbar { justify-content: space-between; flex-wrap: wrap; margin-top: 16px; gap: 16px; }
.toolbar-actions { gap: 12px; }

/* Breadcrumb */
.breadcrumb { gap: 8px; font-size: 0.875rem; color: var(--text-muted); margin-bottom: 12px; }
.breadcrumb a { color: var(--text-muted); text-decoration: none; gap: 6px; }
.breadcrumb a:hover { color: var(--primary); }

/* Button styles */
.btn-secondary {
  background: white;
  border: 1px solid var(--border-color);
  color: var(--text-main);
  padding: 8px 12px;
```

```
}
.btn-secondary:hover { background: #F3F4F6; color:
var(--primary); }
.btn-primary { background: var(--primary); color:
white; border: none; padding: 10px 20px; }
.btn-primary:hover { background: var(--primary-hover);
}
.btn-primary svg { width: 20px; height: 20px; }

/* Product Grid */
.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill,
minmax(250px, 1fr));
  gap: 24px;
  margin-bottom: 32px;
}

/* Product Card */
.product-card {
  background: var(--card-bg);
  border: 1px solid var(--border-color);
  border-radius: var(--radius);
  overflow: hidden;
  display: flex;
  flex-direction: column;
  transition: box-shadow 0.2s, transform 0.2s;
  box-shadow: 0 1px 2px 0 rgba(0,0,0,0.05);
}
.product-card:hover {
  box-shadow: 0 4px 6px -1px rgba(0,0,0,0.1);
  transform: translateY(-2px);
}

.card-image {
  height: 220px;
  padding: 24px;
  display: flex;
```

```
    justify-content: center;
    background: white;
}

.card-image img {
    max-height: 100%;
    max-width: 100%;
    object-fit: contain;
    transition: transform 0.3s;
}

.product-card:hover .card-image img { transform:
scale(1.05); }

.card-content {
    padding: 24px;
    display: flex;
    flex-direction: column;
    flex: 1;
}

.badge {
    background: var(--primary-light);
    color: var(--primary);
    font-size: 0.75rem;
    font-weight: 600;
    padding: 2px 10px;
    border-radius: 4px;
    width: fit-content;
    margin-bottom: 12px;
}

.card-title {
    font-size: 1.125rem;
    font-weight: 600;
    color: var(--text-main);
    text-decoration: none;
    margin-bottom: 8px;
    line-height: 1.4;
    display: -webkit-box;
```

```

    -webkit-line-clamp: 2;
    line-clamp: 2;
    -webkit-box-orient: vertical;
    overflow: hidden;
}
.card-title:hover { text-decoration: underline; }

/* Rating */
.rating-row { gap: 8px; margin-bottom: 16px; }
.stars { display: flex; color: var(--rating-star); }
.rating-score { font-size: 0.875rem; font-weight: 600; }
}
.rating-count, .feature { font-size: 0.875rem; color:
var(--text-muted); }
.features-list { display: flex; gap: 12px; margin-
bottom: 20px; list-style: none; padding: 0; }
.feature { gap: 6px; }
.card-footer { margin-top: auto; justify-content:
space-between; }
.price { font-size: 1.5rem; font-weight: 800; }

/* Cart Sidebar */
#cart-sidebar {
    background: white;
    border-left: 1px solid var(--border-color);
    padding: 24px;
    width: 350px;
    position: fixed;
    right: 0;
    top: 0;
    bottom: 0;
    box-shadow: -4px 0 16px rgba(0,0,0,0.05);
    overflow-y: auto;
    transform: translateX(100%);
    transition: transform 0.3s;
    z-index: 50;
    display: flex;
    flex-direction: column;

```

```
}
#cart-sidebar.open { transform: translateX(0); }
#cart-sidebar h2 { margin-bottom: 24px; font-family:
var(--font-serif); }
.empty-msg { color: var(--text-muted); font-style:
italic; }

/* Overlay */
#overlay {
  display: none;
  position: fixed;
  inset: 0;
  background: rgba(0,0,0,0.3);
  z-index: 40;
}
#overlay.show { display: block; }

.cart-list {
  flex: 1;
  overflow-y: auto;
  display: flex;
  flex-direction: column;
  gap: 16px;
  margin-bottom: 24px;
}
.cart-item {
  display: flex;
  align-items: center;
  gap: 12px;
  padding-bottom: 12px;
  border-bottom: 1px solid var(--border-color);
}
.cart-item img {
  width: 50px;
  height: 50px;
  object-fit: cover;
  border-radius: 4px;
  border: 1px solid var(--border-color);
}
```

```
}
.cart-details h4 {
  font-size: 0.875rem;
  font-weight: 600;
  margin-bottom: 4px;
  display: -webkit-box;
  -webkit-line-clamp: 1;
  line-clamp: 1;
  -webkit-box-orient: vertical;
  overflow: hidden;
}

.cart-details span { font-size: 0.875rem; color: var(--text-muted); }

.cart-footer-summary {
  border-top: 2px solid var(--border-color);
  padding-top: 20px;
  margin-top: auto;
}

.total-row {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 16px;
  font-weight: 700;
  font-size: 1.125rem;
}

.total-price { color: var(--primary); }
#checkout-btn { width: 100%; }

/* Newsletter Section */
.newsletter-section {
  margin-top: 64px;
  background: var(--card-bg);
  border-top: 1px solid var(--border-color);
  padding: 48px 0;
  text-align: center;
}
```



```
.newsletter-content { max-width: 500px; margin: 0 auto;
}
.newsletter-content h3 {
  font-family: var(--font-serif);
  font-size: 1.5rem;
  margin-bottom: 8px;
}
.newsletter-content p { color: var(--text-muted);
margin-bottom: 24px; }
.newsletter-content form { display: flex; gap: 8px; }
.newsletter-content input {
  flex: 1;
  padding: 10px;
  border: 1px solid var(--border-color);
  border-radius: var(--radius);
  font-size: 0.875rem;
}
.newsletter-content input:focus {
  outline: none;
  border-color: var(--primary);
}

#app { width: 100%; }

/* Cart inline section (bawah produk) */
.cart-section {
  margin-top: 32px;
  background: white;
  border: 1px solid var(--border-color);
  border-radius: var(--radius);
  padding: 24px;
}
.cart-section h2 {
  font-family: var(--font-serif);
  margin-bottom: 16px;
  font-size: 1.25rem;
}
```

```

/* Loading Skeleton */
.skeleton {
  background: linear-gradient(90deg, #f0f0f0 25%,
#e0e0e0 50%, #f0f0f0 75%);
  background-size: 200% 100%;
  animation: skeleton-loading 1.5s infinite;
  border-radius: var(--radius);
}
@keyframes skeleton-loading {
  0% { background-position: 200% 0; }
  100% { background-position: -200% 0; }
}

/* Toast notification */
.toast {
  position: fixed;
  bottom: 24px;
  left: 50%;
  transform: translateX(-50%) translateY(80px);
  background: #111827;
  color: white;
  padding: 12px 24px;
  border-radius: 999px;
  font-size: 0.875rem;
  z-index: 100;
  opacity: 0;
  transition: all 0.3s;
  white-space: nowrap;
}
.toast.show {
  opacity: 1;
  transform: translateX(-50%) translateY(0);
}

```

Core / store.js

```
import { deepClone } from "../utils.js";

let state = {
  products: [],
  cart: [],
};

export const getState = () => deepClone(state);

export const setState = (newState) => {

  const nextState = { ...state, ...newState };
  state = nextState;
  console.log("[Store] State diperbarui:", state);
};
```

Core/ utils.js

```
export function deepClone(obj) {

  if (obj === null || typeof obj !== "object") {
    return obj;
  }

  if (obj instanceof Date) {
    return new Date(obj.getTime());
  }

  if (Array.isArray(obj)) {
    return obj.map((item) => deepClone(item));
  }
}
```

```

const clonedObj = {};
for (const key in obj) {
  if (Object.prototype.hasOwnProperty.call(obj, key))
{
    clonedObj[key] = deepClone(obj[key]);
  }
}

return clonedObj;
}

```

Component/ cart.js

```

/**
 *
 * @param {Array} cart
 * @param {Function} onCheckout
 */
export function Cart(cart, onCheckout) {

  const sidebar = document.getElementById("cart-
sidebar");

  const listEl = document.getElementById("sidebar-cart-
list");
  const footerEl = document.getElementById("sidebar-
cart-footer");

  if (!listEl || !footerEl) return;

  listEl.innerHTML = "";
  footerEl.innerHTML = "";

  if (cart.length === 0) {

```

```
listEl.innerHTML = "<p class='empty-msg'>Keranjang  
masih kosong 🛒</p>";  
return;  
}
```

```
cart.map((item) => {  
  const div = document.createElement("div");  
  div.className = "cart-item";  
  div.innerHTML = `  
      
    <div class="cart-details">  
      <h4>${item.title}</h4>  
      <span>${item.price}</span>  
    </div>  
  `;  
  listEl.appendChild(div);  
});
```

```
const total = cart.reduce((acc, item) => acc +  
(item.price || 0), 0);
```

```
footerEl.innerHTML = `  
  <div class="total-row">  
    <span>Total:</span>  
    <span class="total-  
price">${total.toFixed(2)}</span>  
  </div>  
  <button class="btn-primary" id="checkout-btn"  
style="width:100%;">  
    Buy Now 🛒  
  </button>  
`;
```

```
footerEl.querySelector("#checkout-  
btn").addEventListener("click", onCheckout);
```

```

const inlineCart = document.getElementById("cart");
if (inlineCart) {
  inlineCart.innerHTML = `
    <h2>Your Cart (${cart.length} item${cart.length >
1 ? "s" : ""})</h2>
    ${cart.length === 0
      ? "<p class='empty-msg'>Cart is empty.</p>"
      : `<ul style="list-style:none; padding:0;">
        ${cart
          .map(
            (item) =>
              `<li style="display:flex; align-
items:center; gap:12px; padding:8px 0; border-
bottom:1px solid var(--border-color);">
                
                <span>${item.title}</span>
                <span style="margin-left:auto;
font-weight:700; color:var(--
primary);">${item.price}</span>
              </li>`
          )
        }.join("")}
      </ul>
      <p style="margin-top:12px; font-
weight:700;">
        Total: <span style="color:var(--
primary);">${total.toFixed(2)}</span>
      </p>`
  ;
}
}

```

```

export function Newsletter() {

  const $container = $('<div class="newsletter-
section"></div>');

  const $content = $(`
    <div class="newsletter-content">
      <h3>Subscribe to Our Updates</h3>
      <p>Dapatkan berita teknologi dan penawaran
terbaru langsung ke email Anda.</p>
      <form id="newsletter-form">
        <input type="email" placeholder="Masukkan email
Anda" required />
        <button type="submit" class="btn-
primary">Subscribe</button>
      </form>
      <div class="success-message" style="display:none;
margin-top:16px; color:#0D9488; font-weight:600;">
        ☒ Terima kasih telah subscribe!
      </div>
    </div>
  `);

  $container.append($content);
  $("#app").append($container);

  $content.find("form").on("submit", function (e) {
    e.preventDefault();
    const $form = $(this);

    $form.slideUp(400, function () {
      $content.find(".success-message").fadeIn(300);
    });
  });
}

```

Productlist.js

```
function getStarSVG() {
  return `<svg xmlns="http://www.w3.org/2000/svg"
fill="currentColor" viewBox="0 0 20 20">
  <path d="M9.049 2.927c.3-.921 1.603-.921 1.902
0l1.07 3.292a1 1 0
  00.95.69h3.462c.969 0 1.371 1.24.588 1.81l-2.8
2.034a1 1 0 00-.364 1.118l1.07 3.292c.3.921-.755 1.688-
1.54 1.118l-2.8-2.034a1 1 0 00-1.175 0l-2.8 2.034c-
.784.57-1.838-.197-1.539-1.118l1.07-3.292a1 1 0 00-
.364-1.118L2.98 8.72c-.783-.57-.38-1.81.588-
1.81h3.461a1 1 0 00.951-.69l1.07-3.292z">
  </path>
</svg>`;
}

/**
 *
 * @param {Array} products
 * @param {Function} onAdd
 */
export function ProductList(products, onAdd) {
  const container = document.getElementById("product-
list");
  container.innerHTML = "";

  products.map((product) => {
    const card = document.createElement("div");
    card.className = "product-card";

    const { title, price, image, rating = { rate: 4.5,
count: 100 } } = product;
    const discount = Math.floor(Math.random() * 20) +
5;
```



```

card.innerHTML = `
  <div class="card-image">
    
  </div>
  <div class="card-content">
    <span class="badge">Up to ${discount}%
off</span>
    <a href="#" class="card-title"
title="${title}">${title}</a>

    <div class="rating-row">
      <div class="stars">
        ${getStarSVG().repeat(5)}
      </div>
      <span class="rating-
score">${rating.rate}</span>
      <span class="rating-
count">(${rating.count})</span>
    </div>

    <ul class="features-list">
      <li class="feature">
        <svg xmlns="http://www.w3.org/2000/svg"
fill="none" viewBox="0 0 24 24">
          <path stroke="currentColor" stroke-
linecap="round" stroke-linejoin="round"
stroke-width="2" d="M13 7h6l2 4m-8-4v8m0-
8V6a1 1 0 0 0-1-1H4a1 1 0 0 0-1
1v9h2m8 0H9m4 0h2m4 0h2v-4m0 0h-5m3.5
5.5a2.5 2.5 0 1 1-5 0 2.5 2.5 0 0 1
5 0Zm-10 0a2.5 2.5 0 1 1-5 0 2.5 2.5 0 0
1 5 0Z" />
          </svg>
          Fast Delivery
        </li>
      </ul>

```

```

        <div class="card-footer">
            <span class="price">${price}</span>
            <button class="add-btn btn-primary">
                <svg xmlns="http://www.w3.org/2000/svg"
fill="none" viewBox="0 0 24 24">
                    <path stroke="currentColor" stroke-
linecap="round" stroke-join="round"
                        stroke-width="2" d="M4 4h1.5L8 16m0 0h8m-
8 0a2 2 0 1 0 0 4 2 2 0 0 0
                        0-4Zm8 0a2 2 0 1 0 0 4 2 2 0 0 0 0-
4Zm.75-3H7.5M11 7H6.312M17 4v6m-3-3h6" />
                    </svg>
                    Add
                </button>
            </div>
        </div>
    `;

    card.querySelector(".add-
btn").addEventListener("click", function () {
        onAdd(product);
    });

    container.appendChild(card);
});
}

```

API/ productAPI.js

```

/**
 *
 * @returns {Promise<Array>} - array produk
 */
export const fetchProducts = async () => {

```

```
return new Promise((resolve) => {
  // Simulasi network delay (800ms) seperti fetch
  dari server
  setTimeout(() => {
    resolve([
      {
        id: 1,
        title: "Mechanical Keyboard",
        price: 150,
        rating: { rate: 4.8, count: 230 },
        image:
          "https://images.unsplash.com/photo-1618384887929-16ec33fab9ef?auto=format&fit=crop&w=800&q=80",
      },
      {
        id: 2,
        title: "Gaming Mouse",
        price: 80,
        rating: { rate: 4.6, count: 150 },
        image:
          "https://images.unsplash.com/photo-1527814050087-3793815479db?auto=format&fit=crop&w=800&q=80",
      },
      {
        id: 3,
        title: '4K Monitor 24"',
        price: 300,
        rating: { rate: 4.9, count: 85 },
        image:
          "https://images.unsplash.com/photo-1527443224154-c4a3942d3acf?auto=format&fit=crop&w=800&q=80",
      },
      {
        id: 4,
        title: "Premium Coffee Beans",
```

```

        price: 25,
        rating: { rate: 4.7, count: 500 },
        image:
            "https://images.unsplash.com/photo-
1559056199-
641a0ac8b55e?auto=format&fit=crop&w=800&q=80",
    },
    {
        id: 5,
        title: "Travel Tumbler",
        price: 35,
        rating: { rate: 4.5, count: 120 },
        image:
            "https://images.unsplash.com/photo-
1578500494198-
246f612d3b3d?auto=format&fit=crop&w=800&q=80",
    },
    {
        id: 6,
        title: "Noise Cancelling Headphones",
        price: 200,
        rating: { rate: 4.8, count: 320 },
        image:
            "https://images.unsplash.com/photo-
1505740420928-
5e560c06d30e?auto=format&fit=crop&w=800&q=80",
    },
  ]);
}, 800);
});
};

```

Analisis dan Pembahasan

Praktikum ini membahas pembuatan aplikasi **Vanilla JS Web Store** berbasis JavaScript ES6+ dengan struktur modular. Aplikasi menampilkan daftar produk, fitur keranjang belanja, checkout, notifikasi toast, skeleton loading, serta form newsletter. Struktur aplikasi sudah dipisahkan ke beberapa modul, seperti productApi.js untuk simulasi

pengambilan data produk, Store.js untuk manajemen state, ProductList.js untuk menampilkan produk, Cart.js untuk keranjang belanja, dan Newsletter.js untuk fitur langganan email. Pemisahan ini membuat kode lebih rapi, mudah dipahami, dan lebih mudah dikembangkan.

Aplikasi juga telah menerapkan konsep **asynchronous JavaScript** melalui penggunaan `async/await` saat mengambil data produk. Selain itu, terdapat penggunaan DOM manipulation, event handling, array method seperti `map`, `find`, dan `reduce`, serta penggunaan jQuery untuk beberapa interaksi seperti toast notification, sidebar cart, dan newsletter.

Pada saat aplikasi dijalankan, halaman utama menampilkan daftar produk dalam bentuk card yang berisi gambar, nama produk, harga, rating, diskon, dan tombol **Add**. Data produk diperoleh dari fungsi `fetchProducts()` yang mensimulasikan proses pengambilan data dari server dengan delay 800 ms. Selama data belum tampil, aplikasi menampilkan skeleton loading agar tampilan lebih interaktif dan tidak terlihat kosong. Fitur keranjang belanja bekerja dengan cara menyimpan produk yang dipilih ke dalam state. Saat tombol **Add** diklik, sistem akan memeriksa apakah produk sudah ada di keranjang. Jika sudah ada, pengguna akan mendapat notifikasi bahwa produk tersebut telah ditambahkan sebelumnya. Jika belum ada, produk masuk ke cart, jumlah item pada badge diperbarui, tampilan dirender ulang, dan muncul toast notification.

Komponen Cart menampilkan isi keranjang baik pada sidebar maupun bagian bawah halaman. Total harga dihitung menggunakan `reduce`, kemudian ditampilkan dalam format dua angka desimal. Tombol **Buy Now** menjalankan fungsi `checkout` yang mengosongkan keranjang, menutup sidebar, memperbarui jumlah cart, dan menampilkan pesan pembelian berhasil.

Dari sisi tampilan, file CSS sudah mengatur desain modern dengan penggunaan grid responsif, card produk, hover effect, sidebar cart, overlay, skeleton loading, toast notification, dan newsletter section. Hal ini membuat aplikasi lebih menarik dan nyaman digunakan.

Secara keseluruhan, praktikum ini menunjukkan penerapan konsep penting dalam JavaScript modern, yaitu modularisasi, state management sederhana, manipulasi DOM, event listener, asynchronous programming, dan integrasi jQuery. Namun, fitur filter dan sort masih berupa placeholder sehingga belum berfungsi penuh. Selain itu, data produk masih bersifat statis dari simulasi API, belum benar-benar terhubung ke backend atau database.

Kesimpulan

Berdasarkan praktikum ini, dapat disimpulkan bahwa aplikasi Vanilla JS Web Store berhasil dibuat dengan menerapkan konsep JavaScript ES6+ secara modular dan interaktif. Aplikasi mampu menampilkan produk, menambahkan produk ke keranjang, menghitung total harga, melakukan checkout, menampilkan notifikasi, serta menyediakan fitur newsletter. Praktikum ini membantu memahami cara membangun aplikasi web sederhana tanpa framework besar, tetapi tetap terstruktur, responsif, dan mudah dikembangkan. Untuk pengembangan selanjutnya, fitur filter, sort, penghapusan item cart, dan integrasi API nyata dapat ditambahkan agar aplikasi menjadi lebih lengkap.